RAJALAKSHMI ENGINEERING COLLEGE

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P2

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|---|---|
| **Started on** | Thursday, 20 August 2026, 10:22 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 10:42 AM |
| **Time taken** | 20 mins 3 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

void quickSort(int a[], int low, int high) {
    if (low < high) {
        int pivot = a[high];
        int i = low - 1;
        int j, temp;

        for (j = low; j < high; j++) {
            if (a[j] < pivot) {
                i++;
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }

        temp = a[i + 1];
        a[i + 1] = a[high];
        a[high] = temp;

        int p = i + 1;

        quickSort(a, low, p - 1);
        quickSort(a, p + 1, high);
    }
}

int main() {
    int n, i;
    int a[100];

    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    quickSort(a, 0, n - 1);

    for (i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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